

Working with Data

Session 3 — Can I trust the answer?

Block 5 — Pipelines: bronze, silver, gold — cleaning as code

Knowledge check 3 — 10 minutes, closed book, 3 questions

- Covers **Session 2 and Homework 2**: joins, grain, the counts, JSON into tables
- One scenario, three questions. Each can be answered on its own
- One question is a query you write by hand. It is scored clause by clause
- Closed book means closed: no laptop, no phone, no notes
- **This block's lecture is 35 minutes**, not 45: the check comes out of the lecture, never out of the lab

This block in one sentence

Raw data is never edited. Every layer is made by code from the layer below, and every run makes the same tables.

Bronze is what arrived. Silver is what you can trust. Gold is the answer.

And today's version of the course's thesis: a cleaning step that runs without an error has not shown that it kept every value.

Nothing is typed in this lecture. On a prediction slide you write your answer and one reason, in silence; we vote; then the next slide shows what it printed.

Bronze, silver, gold

every layer is made by code from the layer below; every run makes the same tables



Watch — look before you load

The **lecture file** is `data/raw/env_air_gge_ghg.tsv`, greenhouse-gas emissions. (The Lab 5 project has a second file: that one is the lab's.)

```
$ head -n 3 data/raw/env_air_gge_ghg.tsv | cut -c 1-72
freq,unit,airpol,src_crf,geo\TIME_PERIOD      2010      2011      2012      2013      2014      2
A,MIO_T,GHG,CRF1,AT      59.33986      57.11387      54.89359      55.09796      51.36292      53
A,MIO_T,GHG,CRF1,BE      99.26363      89.79709      88.92666      88.69095      82.7348      86.
$ grep 'CRF1B1,AT' data/raw/env_air_gge_ghg.tsv | cut -c 1-60
A,MIO_T,GHG,CRF1B1,AT      0 d      0 d      0 d      0 d      0 d      0 d      0 d      0 d      0 d      0
A,THS_T,GHG,CRF1B1,AT      0 d      0 d      0 d      0 d      0 d      0 d      0 d      0 d      0 d      0
```

- The first column packs five fields into one: **freq,unit,airpol,src_crf,geo\TIME_PERIOD**
- Each **year is a column**. Each cell is **text**: a number, a number and a flag, or **:**

Predict, then watch — bronze: the file, as text

```
con.sql("""
    CREATE OR REPLACE VIEW bronze AS
    SELECT * FROM read_csv('data/raw/env_air_gge_ghg.tsv',
                           delim = '\t', all_varchar = true, names = ['series'])
""")
# and without all_varchar: what type does DuckDB guess for the year columns?
con.sql("DESCRIBE SELECT * FROM read_csv('data/raw/env_air_gge_ghg.tsv', delim = '\t')")
```

`all_varchar = true`: every cell arrives as text. `names = ['series']`: a short name for the packed first column.

Predict: **without** `all_varchar`, DuckDB guesses a type for each column. Would it read the year columns as numbers, or as text? Why?

Bronze: the file, as text — what it printed

```
cell 2 con.sql("""
    CREATE OR REPLACE VIEW bronze AS
    SELECT * FROM read_csv('data/raw/env_air_gge_ghg.tsv',
                           delim = '\t', all_varchar = true, names = ['series'])
    """)
# The prediction: WITHOUT all_varchar, what type would DuckDB guess for the year columns?
con.sql("""
    SELECT column_name, column_type
    FROM (DESCRIBE SELECT * FROM read_csv('data/raw/env_air_gge_ghg.tsv', delim = '\t'))
    WHERE column_name IN ('2010', '2022', '2024')
    """).df()
```

	column_name	column_type
0	2010	VARCHAR
1	2022	VARCHAR
2	2024	VARCHAR

Watch — long: one observation per series and year

cell 3

```
con.sql("""
CREATE OR REPLACE VIEW long AS
SELECT split_part(series, ',', 1) AS freq,
       split_part(series, ',', 2) AS unit,
       split_part(series, ',', 3) AS airpol,
       split_part(series, ',', 4) AS src_crf,
       split_part(series, ',', 5) AS geo,
       CAST(year AS INTEGER)      AS year,
       cell
FROM read_csv('data/raw/env_air_gge_ghg.tsv', delim = '\t', all_varchar = true, names = ['series'])
UNPIVOT INCLUDE NULLS (cell FOR year IN (COLUMNS('^([0-9]{4})$')))
""")
con.sql("SELECT COUNT(*) AS observations FROM long").df()
```

	observations
0	156660

Bronze: one row per series, the years across. Long: one row per observation, one series in one year.

INCLUDE NULLS : an empty cell stays a row.

Watch — the census of the flag

```
cell 4 con.sql("""  
    SELECT split_part(cell, ' ', 2) AS flag, COUNT(*) AS n  
    FROM long  
    GROUP BY flag  
    ORDER BY n DESC  
""").df()
```

	flag	n
0		117530
1	m	38798
2	d	304
3	@C	28

The census first, always: before you convert a column, read every value in it.

Predict, then watch — every cell that is not a number

```
con.sql("SELECT cell, COUNT(*) AS n FROM long WHERE cell LIKE ':%' GROUP BY cell ORDER BY n DESC").df()
```

`:` means "no number here".

Predict: is a `:` cell one kind of cell, or several? Why?

Every cell that is not a number — what it printed

cell 5

```
con.sql("""  
    SELECT cell, COUNT(*) AS n  
    FROM long  
    WHERE cell LIKE ':%'  
    GROUP BY cell  
    ORDER BY n DESC  
""").df()
```

	cell	n
0	: m	38798
1	: d	90
2	:	86
3	: @C	28

Predict, then watch — **CAST**: numbers, or an error?

```
con.sql("""  
    SELECT geo, src_crf, year, CAST(cell AS DOUBLE) AS value  
    FROM long  
    WHERE unit = 'MIO_T'  
""").df()
```

CAST(x AS DOUBLE) turns text into a number.

Predict: a column of numbers, or an error? Why?

CAST — what it printed

cell 6a

```
try:
    con.sql("""
        SELECT geo, src_crf, year, CAST(cell AS DOUBLE) AS value
        FROM long
        WHERE unit = 'MIO_T'
        """).df()
except duckdb.Error as e:
    print("DuckDB refused:", str(e).splitlines()[0])
```

DuckDB refused: Conversion Error: Could not convert string ': m' to DOUBLE when casting from source column cell

Predict, then watch — a first silver: one unit, one cast

```
con.sql("""
    CREATE OR REPLACE VIEW silver_first AS
    SELECT geo, src_crf, year, TRY_CAST(cell AS DOUBLE) AS value
    FROM long
    WHERE unit = 'MIO_T'
""")
con.sql("SELECT COUNT(*) AS rows, COUNT(value) AS with_a_value FROM silver_first").df()
```

`TRY_CAST` is `CAST` that never errors: text it cannot read becomes `NULL`, silently. This runs with no error, and it keeps one unit of two: half the rows of `long`.

Predict: did every number Eurostat wrote survive the cast? Yes or no, and why?

Your Lab 5 skeleton contains a parser like this one, for the waste file.

A first silver: one unit, one cast — what it printed

cell 6b

```
con.sql("""
    CREATE OR REPLACE VIEW silver_first AS
    SELECT geo, src_crf, year, TRY_CAST(cell AS DOUBLE) AS value
    FROM long
    WHERE unit = 'MIO_T'
""")
con.sql("SELECT COUNT(*) AS rows, COUNT(value) AS with_a_value FROM silver_first").df()
```

	rows	with_a_value
0	78330	58722

Watch — what did the cast swallow?

```
cell 7 con.sql("""
    SELECT cell, COUNT(*) AS n
    FROM long
    WHERE TRY_CAST(cell AS DOUBLE) IS NULL      -- the cast gave nothing
          AND split_part(cell, ' ', 1) <> ':'    -- but Eurostat wrote a number
    GROUP BY cell
""").df()
```

	cell	n
0	0 d	214

The answer to the last prediction: the cells where the cast gave nothing, and the source gave something.

Predict, then watch — zeros, and the naive cast

Solid-fuel fugitive emissions (CRF1B1), 2022, the 31 countries. Count the countries with zero emissions, and average them, once with the naive cast and once with the number part of the cell:

```
COUNT(*) FILTER (WHERE TRY_CAST(cell AS DOUBLE) = 0) AS zero_naive_cast,  
COUNT(*) FILTER (WHERE TRY_CAST(split_part(cell, ' ', 1) AS DOUBLE) = 0) AS zero_in_the_file
```

Predict: with the naive cast, does the count of zeros go up, down, or stay the same? And the sum? Why?

The full query: lecture notebook, cell 8.

Zeros, and the naive cast — what it printed

cell 8

```
con.sql("""
    SELECT COUNT(*) FILTER (WHERE TRY_CAST(cell AS DOUBLE) = 0) AS zero_naive_cast,
           COUNT(*) FILTER (WHERE TRY_CAST(split_part(cell, ' ', 1) AS DOUBLE) = 0) AS zero_in_the_file,
           AVG(TRY_CAST(cell AS DOUBLE)) AS average_naive_cast,
           AVG(TRY_CAST(NULLIF(split_part(cell, ' ', 1), ':') AS DOUBLE)) AS average_in_the_file
    FROM long
    WHERE unit = 'MIO_T' AND year = 2022 AND src_crf = 'CRF1B1' AND geo <> 'EU27_2020'
""").df()
```

	zero_naive_cast	zero_in_the_file	average_naive_cast	average_in_the_file
0	11	12	1.032797	0.999481

Watch — the number, the flag, and the cast

```
cell 9 con.sql("""
CREATE OR REPLACE VIEW parsed AS
SELECT *,
        NULLIF(split_part(trim(cell), ' ', 1), ':') AS number_text,
        NULLIF(split_part(trim(cell), ' ', 2), '') AS flag
FROM long
""")
con.sql("""
SELECT COUNT(*) AS swallowed FROM parsed
WHERE number_text IS NOT NULL AND TRY_CAST(number_text AS DOUBLE) IS NULL
""").df()
```

	swallowed
0	0

Three moves: split, **NULLIF** the colon, cast only the number. Then the same count of what the cast swallowed.

Predict, then watch — one country, one year, one sector

Germany's total (T0TX4_MEMO), 2022:

```
SELECT unit, TRY_CAST(cell AS DOUBLE) AS value
FROM long
WHERE geo = 'DE' AND year = 2022 AND src_crf = 'T0TX4_MEMO'
```

Predict: how many rows? And what would SUM(value) of them mean? Why?

The query is cell 10 of the lecture notebook.

One country, one year, one sector — what it printed

```
cell 10 con.sql("""  
    SELECT unit, TRY_CAST(cell AS DOUBLE) AS value  
    FROM long  
    WHERE geo = 'DE' AND year = 2022 AND src_crf = 'TOTX4_MEMO'  
    """).df()
```

	unit	value
0	MIO_T	749.26642
1	THS_T	749266.42000

Watch — the EU among its members

Total greenhouse gases (T0TX4_MEM0), 2022, million tonnes. Three sums: the EU27_2020 row; the 27 member countries; every row.

```
cell 11 con.sql("""
    SELECT SUM(TRY_CAST(cell AS DOUBLE)) FILTER (WHERE geo = 'EU27_2020') AS eu27_row,
           SUM(TRY_CAST(cell AS DOUBLE)) FILTER (
               WHERE geo NOT IN ('EU27_2020', 'CH', 'IS', 'NO', 'TR')) AS sum_of_27_members,
           SUM(TRY_CAST(cell AS DOUBLE)) AS every_row
    FROM long
    WHERE unit = 'MIO_T' AND year = 2022 AND src_crf = 'T0TX4_MEM0'
""").df()
```

	eu27_row	sum_of_27_members	every_row
0	3359.59802	3359.598	7375.81718

Block 1's lesson in different clothes: a **group of countries** sits among the countries. In silver it is kept and marked: **is_aggregate** .

Predict, then watch — every stage has a named key

stage	one row is	key
raw (bronze)	one series, years across	(freq, unit, airpol, src_crf, geo)
long	one observation	(freq, unit, airpol, src_crf, geo, year)
silver — after freq = A, airpol = GHG, one unit	?	?
gold	one country-year, or one country	(geo, year) or geo

```
SELECT COUNT(*), COUNT(DISTINCT (geo, year, src_crf)), COUNT(DISTINCT (geo, year)) FROM silver_first
```

Predict: is (geo, year) the key of silver_first ? Yes or no, and why?

Every stage has a named key — what it printed

cell 12

```
con.sql("""
    SELECT COUNT(*)                                AS rows,
           COUNT(DISTINCT (geo, year, src_crf)) AS geo_year_sector,
           COUNT(DISTINCT (geo, year))          AS geo_year
    FROM silver_first
""").df()
```

	rows	geo_year_sector	geo_year
0	78330	78330	480

The accounting: every observation lands in exactly one place

Counted at the **observation** grain, decided in this order — the first rule that matches wins:

	destination	rule	this file
1	rejected	the cell cannot be read — written to <code>rejects.csv</code> with a reason	0
2	excluded	not the unit silver keeps (<code>THS_T</code>)	78,330
3	aggregate	<code>EU27_2020</code> — kept in silver, <code>is_aggregate</code> = true	2,490
4	retained	everything else — <code>:</code> is retained as <code>NULL</code> , and counted	75,840

156,660 in = 0 + 78,330 + 2,490 + 75,840, printed at the end of every run. With the naive cast, 107 of the retained — the `0 d` zeros in `MIO_T` — would be `NULL` , and nobody would have said so.

Predict, then watch — the precedence, as a **CASE**, counted with **FILTER**

```

CASE
  WHEN cell IS NULL OR trim(cell) = '' THEN 'rejected'
  WHEN number_text IS NOT NULL
        AND TRY_CAST(number_text AS DOUBLE) IS NULL THEN 'rejected'
  WHEN unit <> 'MIO_T' THEN 'excluded'
  WHEN geo = 'EU27_2020' THEN 'aggregate'
  ELSE 'retained'
END AS destination
...
COUNT(*) FILTER (WHERE destination = 'rejected') AS rejected,  -- one column per destination

```

Predict: the five counts come from this one **CASE**. **Could they ever fail to add up** to the observations in? Yes or no, and why?

The full query: lecture notebook, cells 13a (the **CASE**, as a view) and 13b (the counts).

The precedence, as a **CASE**, counted with **FILTER** — what it printed

cell 13b

```
con.sql("""
    SELECT COUNT(*)                                AS observations_in,
           COUNT(*) FILTER (WHERE destination = 'rejected') AS rejected,
           COUNT(*) FILTER (WHERE destination = 'excluded') AS excluded,
           COUNT(*) FILTER (WHERE destination = 'aggregate') AS aggregate,
           COUNT(*) FILTER (WHERE destination = 'retained') AS retained,
           COUNT(*) FILTER (WHERE destination = 'retained' AND number_text IS NULL) AS retained_not_available
    FROM sorted
""").df()
```

	observations_in	rejected	excluded	aggregate	retained	retained_not_available
0	156660	0	78330	2490	75840	19414

The data dictionary is part of silver

column	type	meaning	missing means	made from
geo	VARCHAR	the country (EL Greece) or EU27_2020	never	5th field of series
src_crf	VARCHAR	the source sector (CRF1B1 solid fuels, fugitive)	never	4th field of series
value	DOUBLE	million tonnes of CO ₂ -equivalent	Eurostat wrote :	the number in the cell
flag	VARCHAR	Eurostat's flag: d definition differs, m missing...	no flag	the letters after the number, kept
is_aggregate	BOOLEAN	true for EU27_2020	never	geo

Plus: one row per (geo, year, src_crf) ; one unit, MI0_T . Lab 5 ships one: docs/dictionary.md .

The pandas side note

The same moves, in the words Coding 1 will use:

SQL (this course)	pandas
<code>split_part(cell, ' ', 1)</code>	<code>s.str.split(" ", n=1, expand=True)</code>
<code>TRIM</code> , <code>LOWER</code>	<code>s.str.strip().str.lower()</code>
<code>TRY_CAST(cell AS DOUBLE)</code>	<code>pd.to_numeric(s, errors="coerce")</code> — same swallow: <code>"628 e"</code> → <code>NaN</code> , no error
<code>UNPIVOT</code>	<code>df.melt(...)</code>
<code>strptime</code> / <code>CAST(... AS DATE)</code>	<code>pd.to_datetime(s, format="%Y-%m-%d")</code>

Git thread — raw is committed, generated is ignored

```
# .gitignore (in every pipeline project)
data/silver/
data/gold/
output/
```

- `data/raw/` is committed: it is the evidence, and it never changes
- `data/silver/` , `data/gold/` are **not**: the pipeline makes them, so the pipeline can make them again
- A silver file in `git status` means you edited raw data, or you are about to commit what a script should make
- Commit the fix with the accounting line in the message

AI thread — ask what it could not read

When an assistant cleans a file for you, the code it hands back runs. Ask it three things:

1. What did the cast do to cells that are not just a number?
2. Write the query that counts them.
3. Where do those cells go — silver, a rejects file, or nowhere?

Then run the count yourself. **The assistant cleans what it can see; the accounting line shows what it dropped.**

The class of failure

A conversion that returns `NULL` instead of an error.

- `TRY_CAST`, `pd.to_numeric(errors="coerce")`, a date parse with `try`
- No error. Every row still there. **Values gone.**
- A row count cannot see it. A total may not see it — this file's swallowed values were all zeros.
- Only one thing sees it: **the count of cells where the cast gave nothing and the source gave something.**

5-minute stand-and-stretch — start the clone first

In the **terminal** (Git Bash on Windows, Terminal on macOS), in your course folder, then stand up:

```
git clone https://github.com/earino/ecbs5294-lab05-cleaning.git  
cd ecbs5294-lab05-cleaning  
uv sync
```

It downloads while you stretch. Water, window, hallway. Back in five minutes, then the lab. **The first 20 minutes are AI-off.**

Lab 5 — The clean table that is not clean

Your Lab 5 project: `pipeline.py` runs `scripts/clean.py` (raw → silver) and `scripts/report.py` (silver → gold). The file is the **waste file**, `data/raw/env_wasmun.tsv`.

The report says **15 countries** have a 2024 value for waste generated per person. Germany is blank. Eurostat's file has a number for Germany in 2024.

First minute: check the clone you started ran — VS Code → **File** → **Open Folder...** → `ecbs5294-lab05-cleaning`.

1. `uv run python pipeline.py` — read the report
2. `notebooks/diagnose.ipynb` — look first; **question 5 is yours**, from an empty cell; paste into `DIAGNOSIS.md` before you change anything
3. `scripts/clean.py` — section C, then the four destinations in E; the key check in F after the note

By minute 22: section C rewritten, and a report with more than 15 countries. Not there? Hand up.

AI-off for the first 20 minutes. No stretch task. The neighbour slide comes up on time, finished or not.

The last ten minutes — explain it to a neighbour

Speaker, about a minute, pointing at the screen, not reading the note: symptom → cause → evidence → change → verification.

Listener asks:

1. Show me the query that proves it.
2. Why was it wrong, not just where?
3. Next year's file uses a flag letter you have not seen. What does your parser do with it, and what tells you?

Swap. Unsure, or you disagree? **Hand up.** Then the answer, for everyone. Before you leave: `DIAGNOSIS.md` to the Lab 5 checkpoint on Moodle.